

GIRRAWEE HIGH SCHOOL

MATHEMATICS

YEAR 11

May, 2004

TEST 2

TIME: 90 minutes

- INSTRUCTIONS:
1. Attempt all questions.
 2. Write your answers on your own paper.
 3. All necessary working must be shown.
 4. Marks will be deducted for careless or badly arranged work.
 5. Questions are NOT of equal value.
 6. Start each question on a new page.
 7. Total marks – 120
-

Question 1 (16 marks)

- (a) Evaluate $(3.2)^{25} \div 0.015$ expressing your answer in scientific notation
correct to 3 significant figures. /2
- (b) Evaluate $\sqrt[3]{\frac{9.4 + 3.8}{6.5 - 1.2 \times 3.7}}$ correct to 2 decimal places. /2
- (c) If $\sqrt{5x} = 5\sqrt{2}$, find x . /2
- (d) Write 45g as a percentage of 2.5kg. /1
- (e) Expand and simplify: /2
- (i) $4ab(a - 2b) - 2a^2(b - 3a)$ /2
 - (ii) $(y^2 - 2)(y^2 + 2)$ /1
- (f) Simplify, giving exact answers: /2
- (i) $\sqrt{48} + \sqrt{27}$ /2
 - (ii) $(\sqrt{5} + 2\sqrt{3})^2$ /2
- (g) Simplify $\frac{3}{\sqrt{5}-1}$ by rationalising the denominator. /2

Question 2 (24 marks)

(a) Factorise:

(i) $8 - 18x^2$

(ii) $x^2 - 7x + 12$

(iii) $3x^2 - 20x - 7$

/6

(b) Factorise and simplify:

(i) $\frac{a^2 - 16}{a + 4}$

(ii) $\frac{m^2 - 10m + 25}{m^2 - 2m - 15}$

(iii) $\frac{9 - t^2}{t - 3}$

/7

(c) Simplify:

(i) $\frac{4}{y-2} \div \frac{6}{y-2}$

(ii) $\frac{4m-12}{m^2-9} \times \frac{3m+9}{6m+12}$

/5

(d) Simplify:

(i) $\frac{5-u}{3} - \frac{u+2}{2}$

(ii) $\frac{5}{x-2} - \frac{4}{x+2}$

/6

Question 3 (25 marks)

(a) Solve the equation:

(i) $3(x-5) = 3 - 2(x-1)$

(ii) $\frac{2x-1}{3} = 1 - \frac{x-2}{5}$

/6

(b) Solve the quadratic equation $6x^2 = 4x + 2$ (leaving answer in simplified surd form if necessary). /3(c) The equation $x^2 - ax + 20 = 0$ has one solution $x = 4$.Find a and the other solution.

/3

(d) Solve the simultaneous equations: $2x + y = 8$

$xy = 6$

(e) Solve $2 - x \leq \frac{3x}{4}$ and graph the solution on the number line.

/3

(f) Solve the following:

(i) $|3y + 2| = |6 - 4y|$

(ii) $|2m - 5| > 9$

/6

Question 4 (13 marks)

(a) Find, correct to 3 decimal places:

(i) $\sin 42^{\circ}05'$

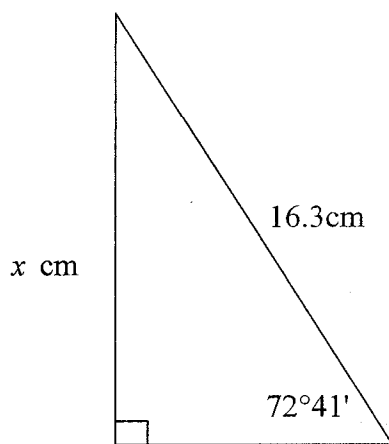
(ii) $\cos 116^{\circ}28'$

/2

(b) If $\tan \theta = \frac{6.1}{8.5}$, find θ to the nearest minute.

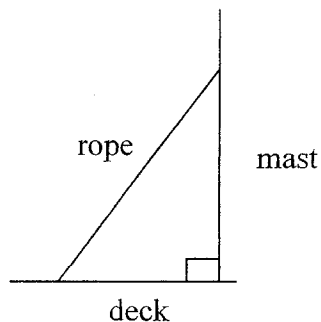
/2

(c) Find the value of x correct to 1 decimal place.



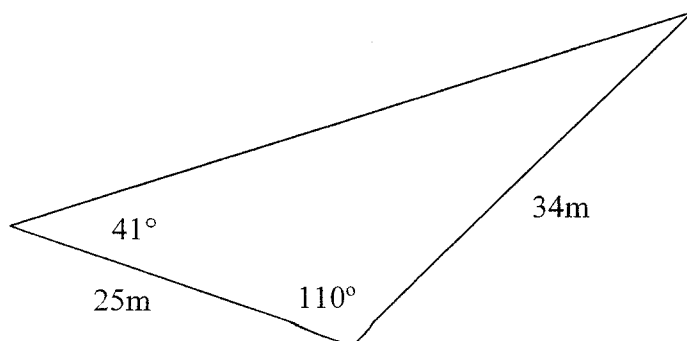
/2

(d) A wire rope is attached to the mast of a boat 3.2m above the deck, and to the deck, making an angle of elevation of $72^{\circ}10'$. How long is the rope?



/3

(e) A triangular block of land is to be made into a park and planted with grass. What is the area to be grassed, to the nearest m^2 ?



/3

Question 5 (23 marks)

(a) Find the exact value of:

(i) $\sin 210^\circ$

(ii) $\cos (-135^\circ)$

(iii) $\cot 300^\circ$

/6

(b) Find the value(s) of θ (to the nearest degree) for $0^\circ < \theta < 360^\circ$ for which:

(i) $\cos \theta = 0.917$

(ii) $\sin \theta = -0.158$

(iii) $\tan \theta = 0.674$ and $\sin \theta$
is negative

/6

(c) Give the exact value, in simplest form:

(i) $\cot 30^\circ + \cot 60^\circ$

(ii) $\frac{\sin 45^\circ + \sin 30^\circ}{\cos 45^\circ - \cos 60^\circ}$

/8

(d) If $\sin \theta = \frac{-5}{13}$ and $\cos \theta < 0$, find the exact value of $\tan \theta$.

/3

Question 6 (20 marks)(a) Solve the following for $0^\circ \leq \theta \leq 360^\circ$

(i) $2\cos \theta = \sqrt{3}$

(ii) $\operatorname{cosec}^2 \theta - 2 = 0$

/7

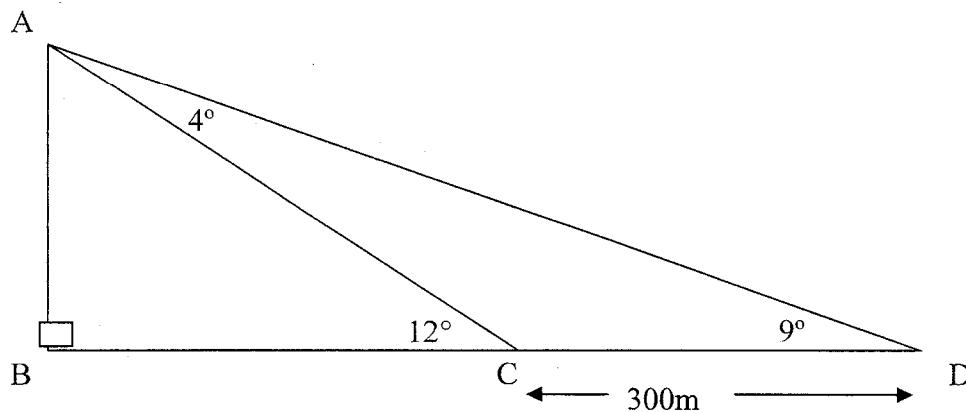
(b) Simplify $\cos^2 \theta (1 + \cot^2 \theta)$.

/2

(c) Prove $\frac{\operatorname{cosec} \theta}{\cot \theta} = \sec \theta$.

/2

(d)



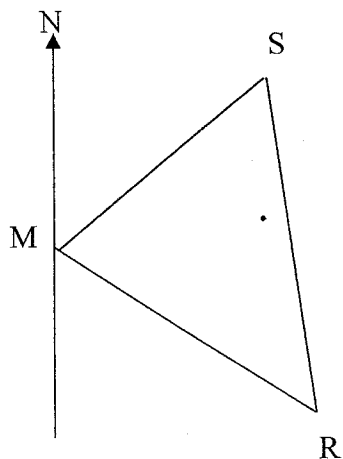
(i) Use the sine rule to calculate AC to the nearest metre.

/2

(ii) Hence, find BC to the nearest metre.

/2

- (e) A boat (S) is sinking 2.2km out to sea from a marina (M). Its bearing is 041° from the marina. A rescue boat (R) is at bearing 123° from the marina and 2.9km out to sea.



- | | | |
|-------|---|----|
| (i) | Copy the diagram and indicate the given information. | /1 |
| (ii) | Show $\angle SMR = 82^\circ$. | /1 |
| (iii) | Use the cosine rule to find the distance RS of the rescue boat from the sinking boat. | /3 |

SOLUTIONS Yr 11 TEST 2

Question 1

18

(a) $2.13 \times 10^{-27} = 2.13 \times 10^{-27}$
(3 sig fig)

(b) $\sqrt[3]{\frac{9.4+3.8}{6.5-1.2 \times 3.7}} = \sqrt[3]{\frac{13.2}{1.15}} = \sqrt[3]{11.477...}$

$= 1.85738... \rightarrow 1.86$ to 2 dec. pl.

(c) $\sqrt{5x} = 5\sqrt{2}$
 $= \sqrt{50}$

$5x = 50$

$x = 10$

(d) $\frac{4g}{2.5kg} = 0.16\%$

(d) (i) $4a^2b - 8ab^2 - 2a^2b + 6a^3$
 $= 6a^3 + 2a^2b - 8ab^2$

(ii) $y^4 - 4$

(e) (i) $\sqrt{48} + \sqrt{27} = \sqrt{16} \times \sqrt{3} + \sqrt{9} \times \sqrt{3}$
 $= 4\sqrt{3} + 3\sqrt{3}$
 $= 7\sqrt{3}$

(ii) $(15)^2 + 2 \times 2 \times \sqrt{15} + (2\sqrt{3})^2$
 $= 5 + 4\sqrt{15} + 4 \times 3$
 $= 17 + 4\sqrt{15}$

(f) $\frac{3}{15-1} \times \frac{15+1}{15+1} = \frac{3(15+1)}{5-1}$
 $= \frac{3 \times 15 + 3}{4}$

Question 2

24

(a) (i) $8 - 18x^2 = 2(4 - 9x^2)$
 $= 2(2+3x)(2-3x)$

(ii) $x^2 - 7x + 12 = (x-3)(x-4)$

(iii) $3x^2 - 20x - 7 = (3x+1)(x-7)$

$3x^2 - 20x - 7 = (3x+1)(x-7)$
 $3x^2 - 20x - 7 = 3x^2 - 21x + x - 7$
 $= 3x(x-7) + 1(x-7)$
 $= (3x+1)(x-7)$

(b) (i) $\frac{a^2-16}{a+4} = \frac{(a+4)(a-4)}{(a+4)}$
 $= a-4$

(iv) $\frac{m^2-10m+25}{m^2-2m-15} = \frac{(m-5)^2}{(m-5)(m+3)}$

$= \frac{m-5}{m+3}$

(iii) $\frac{9-t^2}{t-3} = \frac{(3-t)(3+t)}{(t-3)}$

$= -\frac{(t-3)(3+t)}{(t-3)}$

$= -(3+t)$

$= -t-3$ or $-(t+3)$

5(c)

(i) $\frac{4}{y-2} \div \frac{6}{y-2} = \frac{4}{y-2} \times \frac{y-2}{6}$
 $= \frac{4}{6} = \frac{2}{3}$

$$(c) (ii) \frac{4m-12}{m^2-9} \times \frac{3m+9}{6m+12} = \frac{4(m-3)}{(m+3)(m-3)} \times \frac{3(m+3)}{6(m+2)}$$

$$= \frac{2}{m+2}$$

1/6 (d)

$$(i) \frac{5-u}{3} - \frac{u+2}{2} = \frac{2(5-u) - 3(u+2)}{6}$$

$$= \frac{10-2u-3u-6}{6}$$

$$= \frac{4-5u}{6} \quad 1/3$$

$$(ii) \frac{5(x+2) - 4(x-2)}{(x-2)(x+2)} = \frac{5x+10-4x+8}{x^2-4}$$

$$= \frac{x+18}{x^2-4} \quad 1/3$$

Question 3 1/25

$$(a) (i) 3(x-5) = 3 - 2(x-1)$$

$$3x-15 = 3-2x+2$$

$$= 5-2x$$

$$5x = 20$$

$$x = 4$$

1/3

$$(ii) \frac{2x-1}{3} = 1 - \frac{x-2}{5}$$

$$10x-5 = 15-3x+6$$

$$13x = 26$$

$$x = 2$$

1/3

1/3 (b)

$$6x^2 = 4x+2$$

$$6x^2 - 4x - 2 = 0$$

$$3x^2 - 2x - 1 = 0$$

$$(3x+1)(x-1) = 0$$

$$x = -\frac{1}{3} \text{ or } 1$$

$$\text{or } x = \frac{4 \pm \sqrt{16+48}}{12}$$

$$= \frac{4 \pm \sqrt{64}}{12}$$

$$= \frac{4 \pm 8}{12}$$

$$= \frac{12}{12} \text{ or } \frac{-4}{12}$$

$$= 1 \text{ or } -\frac{1}{3}$$

1/3

1/3

$$(c) x^2 - ax + 20 = 0$$

$$1 \text{ if } x=4, 16-4a+20=0$$

$$36-4a=0$$

$$a=9$$

$$\therefore x^2 - 9x + 20 = 0$$

$$(x-4)(x-5) = 0$$

$$\therefore \text{other sol}^n \text{ is } x=5 \quad 1/2$$

Question 3 cont.

1/4 (d)

$$2x + y = 8 \quad \text{--- (1)}$$

$$xy = 6 \quad \text{--- (2)}$$

$$\text{From (1) } y = 8 - 2x \quad \text{--- (3)}$$

Subst. for y in (2):

$$x(8 - 2x) = 6$$

$$8x - 2x^2 = 6$$

$$2x^2 - 8x + 6 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x-1)(x-3) = 0$$

$$x = 1, 3$$

Subst. for x in (3):

$$y = 8 - 2 \text{ or } 8 - 6$$

$$y = 6 \text{ or } 2$$

$$\therefore \text{Soln } x = 1, y = 6 \text{ or } x = 3, y = 2$$

1/4

1/3 (e) $2 - x \leq \frac{3x}{4}$

$$\therefore 8 - 4x \leq 3x$$

$$8 \leq 7x$$

$$x \geq \frac{8}{7}$$

$$\text{ie } x \geq 1\frac{1}{7}$$

$$\frac{1}{2}$$



$$1$$

1/6 (f)(i) $|3y + 2| = |6 - 4y|$

$$\text{Either } 3y + 2 = 6 - 4y$$

$$7y = 4$$

$$y = \frac{4}{7}$$

$$\text{or } 3y + 2 = 4y - 6$$

$$8 = y$$

$$\therefore \text{Soln } y = \frac{4}{7} \text{ or } 8$$

1/3

(ii) $|2m - 5| > 9$

$$2m - 5 > 9$$

$$2m > 14$$

$$m > 7$$

$$\text{or } -(2m - 5) > 9$$

$$2m - 5 < -9$$

$$2m < -4$$

$$m < -2$$

$$\therefore \text{Soln } m < -2 \text{ or } m > 7$$

1/3

Question 4

13

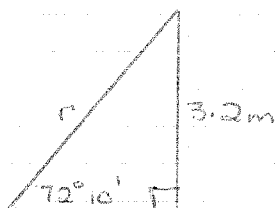
1/2 (a) (i) $\sin 42^\circ 05' = 0.670210...$
 $= 0.670$ to 3 dec.pl. 1

(ii) $\cos 116^\circ 28' = -0.44567...$
 $= -0.446$ to 3 dec.pl. 1

1/2 (b) $\tan \theta = \frac{61}{85}$
 $= 0.717647...$
 $\therefore \theta = 35.665...$
 $= 35^\circ 40'$ 1/2

1/2 (c) $\tan 72^\circ 41' = \frac{x}{16.3}$
 $\therefore x = 16.3 \tan 72^\circ 41'$
 $= 16.3 \times 3.20734...$
 $= 52.2797...$
 $= 52.3$ to 1 dec.pl. 1/2

1/3 (d)



$$\frac{3.2}{r} = \sin 72^\circ 10'$$

$$r = \frac{3.2}{\sin 72^\circ 10'}$$

$$= 3.3615...$$

$$\approx 3.4 \text{ m}$$

16 Rope is 3.4 m

1/3

1/3 (e) Area = $\frac{1}{2} ab \sin C$

$$= \frac{1}{2} \times 25 \times 34 \sin 110^\circ \text{ m}^2$$

$$= 399.369$$

$$= 399 \text{ m}^2 \text{ to nearest m}^2$$

1/3

Question 5

123

16 (a) (i) $\sin 210^\circ = \sin (180^\circ + 30^\circ)$ (ii) $\cos (-135^\circ) = -\cos 45^\circ$

$= -\sin 30^\circ$ $= -\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

(iii) $\cot 300^\circ = -\cot 60^\circ$

$= -\frac{1}{\sqrt{3}}$ $\frac{1}{2}$

16 (b) (i) $\cos \theta = 0.917$

$\theta = 24^\circ$ or $360 - 24^\circ$

$\theta = 24^\circ$ or 336°

(ii) $\sin \theta = -0.158$

$\theta = 180^\circ + 9^\circ$ or $360^\circ + 9^\circ$

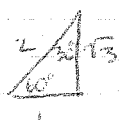
$\theta = 189^\circ$ or 369° $\frac{1}{2}$

(iii) $\tan \theta = 0.674$

$\theta = 180^\circ + 34^\circ$ (3rd quadrant only)

$\theta = 214^\circ$ $\frac{1}{2}$

(c) (i) $\cot 30^\circ + \cot 60^\circ = \sqrt{3} + \frac{1}{\sqrt{3}}$



$= \frac{3+1}{\sqrt{3}}$ or $\frac{\sqrt{3} + \frac{1}{\sqrt{3}}}{1}$

$= \frac{4}{\sqrt{3}}$ $\frac{4\sqrt{3}}{3}$

$= \frac{4\sqrt{3}}{3}$ $\frac{1}{3}$

(ii) $\frac{\sin 45^\circ + \sin 30^\circ}{\cos 45^\circ - \cos 60^\circ} = \frac{\frac{1}{\sqrt{2}} + \frac{1}{2}}{\frac{1}{\sqrt{2}} - \frac{1}{2}}$

$= \left(\frac{\frac{1}{\sqrt{2}} + \frac{1}{2}}{\frac{1}{\sqrt{2}} - \frac{1}{2}} \right) \div \left(\frac{\frac{1}{\sqrt{2}} - \frac{1}{2}}{\frac{1}{\sqrt{2}} - \frac{1}{2}} \right)$

$= \frac{\frac{1}{\sqrt{2}} + 1}{2} \times \frac{2}{\frac{1}{\sqrt{2}} - 1}$

$= \frac{\frac{1}{\sqrt{2}} + 1}{\frac{1}{\sqrt{2}} - 1}$

$= \frac{(\frac{1}{\sqrt{2}} + 1)(\frac{1}{\sqrt{2}} + 1)}{(\frac{1}{\sqrt{2}} - 1)(\frac{1}{\sqrt{2}} + 1)}$

$= \frac{2 + 2\sqrt{2} + 1}{2 - 1}$

$= 3 + 2\sqrt{2}$ $\frac{1}{3}$

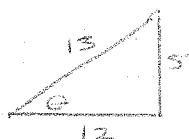
13 (d) $\sin \theta = -\frac{5}{13}$

$\cos \theta < 0$

$\therefore \theta$ in 3rd quadrant

$\therefore \tan \theta > 0$

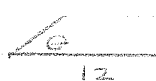
$\tan \theta = \frac{5}{12}$



$13^2 - 5^2 = 144$

$\sqrt{144} = 12$

$\frac{12}{12}$



$\frac{1}{3}$

$\frac{1}{3}$

Question 6

1/4 too much?

(a) (i) $2 \cos \theta = \sqrt{3}$
 $\cos \theta = \frac{\sqrt{3}}{2}$

$\theta = 30^\circ \text{ or } 360^\circ - 30^\circ$
 $= 30^\circ \text{ or } 330^\circ$



(ii) $\operatorname{cosec}^2 \theta - 2 = 0$
 $\operatorname{cosec}^2 \theta = 2$
 $\operatorname{cosec} \theta = \pm \sqrt{2}$
 $\sin \theta = \pm \frac{1}{\sqrt{2}}$

$\theta = 45^\circ, 135^\circ, 225^\circ \text{ or } 315^\circ$

1/2 (b) $\cos^2 \theta (1 + \cot^2 \theta) = \cos^2 \theta \times \operatorname{cosec}^2 \theta$
 $= \frac{\cos^2 \theta}{\sin^2 \theta}$
 $= \cot^2 \theta$

1/2 (c) Pr $\frac{\operatorname{cosec} \theta}{\cot \theta} = \sec \theta$

LHS = $\frac{1}{\sin \theta} \div \frac{\cos \theta}{\sin \theta}$

$= \frac{1}{\sin \theta} \times \frac{\sin \theta}{\cos \theta}$

$= \frac{1}{\cos \theta}$

$= \sec \theta$

$= \text{RHS}$

1/4 (d) (i) In $\triangle ACD$, $\frac{AC}{\sin 9^\circ} = \frac{300}{\sin 4^\circ}$

$\therefore AC = \frac{300 \sin 9^\circ}{\sin 4^\circ}$

$= 672.7 \dots$

$= 673 \text{ m to nearest m}$

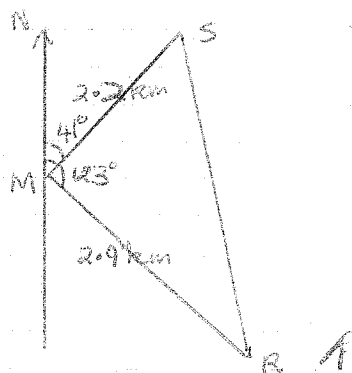
(ii) In $\triangle ABC$, $\frac{BC}{AC} = \cos 12^\circ$

$\therefore BC = 673 \cos 12^\circ$

$= 558.29 \dots$ (or $658.07 \dots$ using all dec. pl.)

$= 558 \text{ m to nearest m}$

1/5 (e) (i)



(ii) $\angle SMR = 123^\circ - 41^\circ$
 $= 82^\circ$

(iii) $RS^2 = MR^2 + MS^2 - 2MRMS \cos 82^\circ$
 $= (2.9)^2 + (2.2)^2 - 2 \times 2.9 \times 2.2 \cos 82^\circ$
 $= 11.47 \dots$

$\therefore RS = 3.387 \dots$

$= 3.4 \text{ km (1 dec. pl.)}$

$\therefore RS = 3.381 \dots$

$= 3.4 \text{ km (1 dec. pl.)}$